

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow engine** for MCA company-registration filings, reducing manual filing time by **85%** through state recovery, retries, and public-site API lookups.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker** and automated **GitHub Actions** deployments to **Google Cloud Run**, improving release velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, turning model outputs into usable masks and 3D visualization artifacts inside an existing application workflow.
- Reduced processing runtime by **87%** with async/parallel execution and fewer repeated post-processing passes over large MRI volumes.
- Integrated outputs into 3D reconstructed surface visualization flows so model results stayed inspectable and useful to end users.

Open Source

[Contributions](#) 

- Contributed to Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working with existing maintainers, code style, and review constraints.

Projects

Distributed Inference Engine | *Python, PyTorch, FastAPI, gRPC, CUDA, Docker, Redis*

- Built a **distributed inference engine** to run models larger than a single consumer GPU by coordinating token generation across heterogeneous nodes with speculative decoding.
- Designed streaming APIs, worker heartbeats, retry-aware orchestration, and KV-cache-aware routing to balance latency, throughput, and worker availability across local and remote inference workers.

Seaweed: Contest Platform | *Go, PostgreSQL, Redis, Judge0, AWS, Docker, Kubernetes*

[GitHub](#) 

- Built backend services for registration, submissions, sandboxed execution, scoring, and live contest rankings across **500+ concurrent users**.
- Used async Go workers, Judge0, PostgreSQL, Redis, and Docker/Kubernetes deployment paths on AWS to keep evaluation workflows recoverable under contest load.

Penny Lane Capital (Tradeage) | *Python, Agno, Pydantic, SQLite, yfinance, JSONL*

- Built an **Agno Workflow** multi-agent research system with analyst roles, bull/bear debate, trader, risk review, and portfolio approval using Pydantic state, checkpoints, memory, and JSONL traces.
- Evaluated agent decisions with historical-market backtests, reward loops, validation harnesses, and **5 baselines**, producing return, drawdown, and win-rate scorecards across market windows.

Technical Skills

Languages: Python, Go, TypeScript, C++

Frameworks: FastAPI, Echo, Node.js, Express, PyTorch, gRPC

Tools: Redis, PostgreSQL, Docker, Prometheus, OpenTelemetry, RabbitMQ